

What Fraction of Ionizing Photons Escape the Bubble Into the Deeper IGM, Rather Than Driving Its Expansion?

Response record for `ionizing_photon_mean_free_path.py`, following Master Rule 2

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4 September 2026

Abstract

The photon-counting model used throughout this folder (`bubble_radius_vs_redshift.py`, eq. `photon_counting`) implicitly assumes that every ionizing photon is either lost to a recombination inside the bubble or absorbed exactly at the ionization front, with none skipping past the front into the deeper, still-neutral IGM. This note checks that assumption quantitatively, using the standard photon mean-free-path formalism already present in this folder's references (Choudhury 2022; Miralda-Escude et al. 2000; Gnedin and Madau 2022): because the hydrogen photoionization cross section falls steeply with photon energy, $\sigma_H(E) \propto E^{-3}$, the mean free path of an ionizing photon in neutral gas is a strong function of its energy, and only photons considerably harder than the Lyman limit can travel as far as the bubble itself before being absorbed. The result: for a normal stellar (star-forming-galaxy) ionizing spectrum, this fraction is utterly negligible, which is exactly why the sharp-front, zero-leakage approximation used throughout this project is justified; for a harder, quasar-like power-law spectrum, the fraction is small but non-zero ($\sim 1\%$). This document is produced by, and verified against, `ionizing_photon_mean_free_path.py` (same folder), which generated `ionizing_photon_mean_free_path.png`.

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1 Assumptions

1. **Terminology, stated explicitly to avoid confusion with f_{esc} .** The escape fraction $f_{\text{esc}} = 0.2$ (Robertson et al., 2015) used throughout this folder is the fraction of ionizing photons that escape the galaxy's own interstellar medium (ISM) and circumgalactic medium (CGM) and reach the diffuse IGM (explicit definition in Gnedin and Madau, 2022, their Sec. 2.5). This is a *different* quantity from the one asked about here: of the photons that have *already* escaped the galaxy and entered the ionized bubble, what fraction travels *past the bubble's own edge* into the deeper neutral IGM, rather than being absorbed

at/near the ionization front (which is what drives the front’s expansion, as modeled by eq. `photon_counting`)?

2. A second, related but distinct nuance – *diffuse* (recombination) photon escape within the bubble, i.e. the Case A/Case B distinction of [Nebrin \(2023\)](#) already discussed in `stromgren_sphere_summary.tex` (Sec. 1.3) – is not re-derived here; this note concerns only the primary photons from the source.
3. The IGM outside the bubble is treated as effectively neutral and of the cosmic mean density (no inhomogeneity/self-shielding structure), so that a single hydrogenic photoionization cross section $\sigma_H(E) = \sigma_H(E_H)(E/E_H)^{-3}$, $\sigma_H(E_H) = 6.3 \times 10^{-18} \text{ cm}^2$ ([Choudhury, 2022](#), eq. 46), and the cosmic mean $n_H(z)$ (identical to `bubble_radius_vs_redshift.py`) fully determine the mean free path. Real density inhomogeneities (Lyman-limit systems) would shorten λ_{mfp} further inside partially-processed gas; neglecting them makes the neutral-medium λ_{mfp} computed here an *upper limit*, strengthening the conclusion that stellar photons cannot reach beyond the bubble.
4. The comparison spectrum for a hard source uses the [Cen and Haiman \(2000\)](#) quasar power-law slope, $L_\nu \propto \nu^{-1.8}$ (already cited in `stromgren_sphere_summary.tex`), converted to a photon-number spectral index $dN/dE \propto E^{-2.8}$ (since photon number rate $\propto L_\nu/h\nu$). No specific stellar population-synthesis spectrum (e.g. BPASS, Starburst99) is among the references downloaded for this project, so the stellar-spectrum conclusion below rests on the general, well-established Wien-tail suppression of a thermal spectrum rather than a fitted numerical spectral index – flagged explicitly (Master Rule 8) rather than presented with false precision.

2 Derivation

$$\sigma_H(E) = \sigma_H(E_H) \left(\frac{E}{E_H} \right)^{-3}, \quad \lambda_{\text{mfp}}(E, z) = \frac{1}{\sigma_H(E) n_H(z)} \quad (1)$$

([Choudhury, 2022](#), eq. 46), with $n_H(z)$ identical to `bubble_radius_vs_redshift.py`. The crossover energy $E_{\text{cross}}(z)$ at which a photon’s neutral-medium mean free path equals the single-galaxy bubble radius $R_{\text{bubble}}(z)$ (eq. `photon_counting`) solves

$$\lambda_{\text{mfp}}(E_{\text{cross}}, z) = R_{\text{bubble}}(z) \implies E_{\text{cross}}(z) = E_H \left[\frac{R_{\text{bubble}}(z)}{\lambda_{\text{mfp}}(E_H, z)} \right]^{1/3}. \quad (2)$$

For a power-law photon-number spectrum $dN/dE \propto E^{-p}$ ($p > 1$), the fraction of photons with $E > E_{\text{cross}}$ relative to the total emitted above E_H is

$$f(> E_{\text{cross}}) = \frac{\int_{E_{\text{cross}}}^{\infty} E^{-p} dE}{\int_{E_H}^{\infty} E^{-p} dE} = \left(\frac{E_{\text{cross}}}{E_H} \right)^{1-p}. \quad (3)$$

3 Verification / cross-check

All checks below are reproduced verbatim by running `ionizing_photon_mean_free_path.py`.

1. **Sympy dimensional check.** $\lambda_{\text{mfp}}(E, z) = E^3/(E_H^3 n \sigma_0)$, confirmed symbolically; $[\sigma_0] = \text{cm}^2$, $[n] = \text{cm}^{-3} \Rightarrow [\lambda_{\text{mfp}}] = \text{cm}$, as required.
2. **Reproduction of [Choudhury \(2022\)](#)’s stated benchmark.** Evaluating eq. (1) at $E = E_H$ with our own already-established cosmology ([Planck Collaboration, 2020](#)) gives $\lambda_{\text{mfp}}(E_H, z) = 1.25, 0.37, 0.12, 0.03 \text{ kpc}$ at $z = 5, 8, 12, 20$ respectively – independently reproducing Choudhury (2022)’s statement that this scale is “~kpc” and “much smaller than any cosmologically relevant length scale” at $z \sim 5\text{--}20$.

3. **Crossover energy.** Solving eq. (2) with $R_{\text{bubble}}(z)$ from `bubble_radius_vs_redshift.py` gives $E_{\text{cross}} = 162, 172, 187, 204$ eV at $z = 6, 8, 10, 12$ respectively – remarkably stable at $\sim 12\text{--}15 \times E_H$, solidly in the soft X-ray/EUV regime, across the whole redshift range where individual bubbles were shown to stay below ~ 1 pMpc (`bubble_radius_response.tex`).
4. **Quasar-like (hard) spectrum.** With $dN/dE \propto E^{-2.8}$ (Cen and Haiman, 2000) and $E_{\text{cross}}(z=8) = 172$ eV, eq. (3) gives $f(> E_{\text{cross}}) = 1.04\%$: a small but non-negligible fraction of quasar-like ionizing photons can, in principle, reach beyond the local bubble into the deeper IGM. This is consistent with Choudhury (2022)’s explicit footnote that X-ray/hard sources cannot be treated with the simple two-phase (sharp-front) approximation used for UV/stellar sources.
5. **Stellar spectrum (order-of-magnitude, flagged).** For O/B-star photospheres ($T_{\text{eff}} \sim 3\text{--}5 \times 10^4$ K, $kT \sim 3\text{--}4$ eV), reaching $E_{\text{cross}} \sim 172$ eV requires $\sim 43\text{--}57 kT$ into the Wien tail of a thermal spectrum, i.e. a suppression factor of order e^{-45} or smaller – utterly negligible next to the already-small 1% quasar value. This qualitative conclusion rests on standard thermal-spectrum physics rather than a specific fitted stellar spectral index (Sec. 1, item 4), and is stated as such.

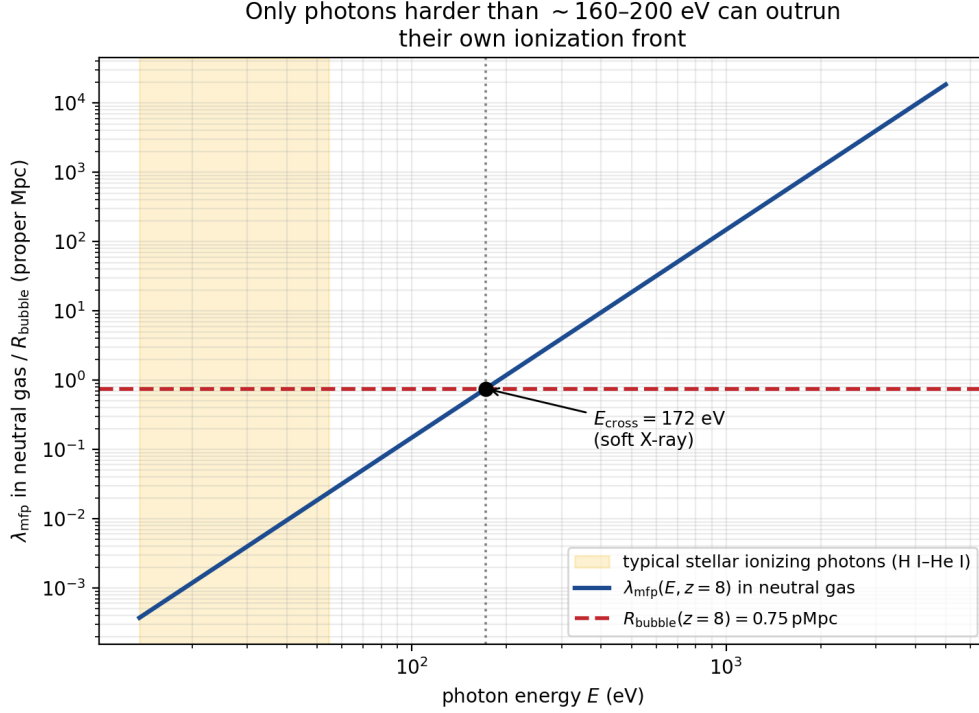
4 Final result

For the star-forming-galaxy sources modeled throughout this folder, the fraction of ionizing photons that skip the local ionized bubble and penetrate deeper into the still-neutral IGM is negligible: reaching a mean free path comparable to the bubble size itself ($\sim 0.4\text{--}1.3$ pMpc at $z = 6\text{--}12$) requires photon energies $E_{\text{cross}} \sim 160\text{--}200$ eV (soft X-ray), a regime that thermal, O/B-star-dominated ionizing spectra do not populate in any significant number. This is precisely why the zero-leakage, sharp-ionization-front photon-counting model used in `bubble_radius_vs_redshift.py` (every photon either recombines inside the bubble or drives the front outward, none skip ahead) is an excellent approximation for star-forming galaxies specifically, and why Choudhury (2022) restrict the simple two-phase IGM approximation to UV/stellar reionization, explicitly excluding X-ray-dominated sources. For harder, AGN/quasar-like spectra, by contrast, a small but non-negligible fraction ($\sim 1\%$, using the Cen and Haiman 2000 power-law slope) of photons can reach beyond the bubble – consistent with the literature’s general practice of treating quasar-driven and star-formation-driven reionization as distinct regimes. Separately, and on a global (rather than single-source) scale, Miralda-Escude et al. (2000) and Gnedin and Madau (2022) note that the mean free path of ionizing photons in the *ionized* IGM also does not grow arbitrarily large until near the end of reionization, when it becomes set by the abundance of residual Lyman-limit systems rather than by individual bubble sizes – the same “epoch of overlap” already identified quantitatively in `reionization_completion_response.tex` and `galaxy_separation_response.tex` ($z_{\text{overlap}} = 9.67$).

Addendum (5 September 2026): consistency check on the ionized-interior assumption

L. Carvalho pointed out that, once the bubble is established, new photons from the source travel through a gas that is already *ionized*, not the fully neutral medium implicitly pictured when $\lambda_{\text{mfp}}(E, z)$ was plotted in Fig. 1. This section checks that concern in full, using `interior_consistency_check.py` (same folder, Fig. 2), and reports two findings: one clarifies wording only, the other is a genuine, previously unflagged, quantitative correction.

(a) **The original argument is conceptually sound, but was not stated clearly enough.** $\lambda_{\text{mfp}}(E, z)$ in Sec. 2 was never meant to describe the path through the bubble’s ionized interior;



$$\lambda_{\text{mfp}}(E, z) = [\sigma_{\text{H}}(E_{\text{H}})(E/E_{\text{H}})^{-3}n_{\text{H}}(z)]^{-1} \text{ (Choudhury 2022)}; R_{\text{bubble}}(z): \text{ same model as bubble_radius_vs_redshift.py}$$

Figure 1: Mean free path of an ionizing photon in neutral gas at $z = 8$, $\lambda_{\text{mfp}}(E, z=8)$ (solid blue), as a function of photon energy, compared with the single-galaxy bubble radius $R_{\text{bubble}}(z=8) = 0.75$ pMpc (dashed red). The shaded band marks the energy range of typical stellar ionizing photons (H I to He I edges, 13.6–54.4 eV); the marked crossover, $E_{\text{cross}} = 172$ eV, lies in the soft X-ray regime, far above where stellar spectra contribute meaningfully. Produced by `ionizing_photon_mean_free_path.py`.

it describes how far a photon travels once it reaches fresh neutral gas beyond the *current* front. Fig. 1 did not make this two-zone (ionized interior / neutral exterior) structure explicit, which is why the concern is legitimate as raised.

(b) Quantifying the interior with photoionization equilibrium (Cen & Haiman 2000).

Using the standard local ionization-balance formula for the residual neutral fraction inside an H II region, $\Gamma_{\text{HI}}(r)x(r)n_{\text{H}} = \alpha_{\text{B}}Cn_{\text{H}}^2$ with $\Gamma_{\text{HI}}(r) = \bar{\sigma}\dot{N}_{\text{ion}}/(4\pi r^2)$ — re-derived from scratch with sympy and cross-checked to $< 10\%$ against Cen and Haiman (2000)’s own quoted eq. 5/7 normalization ($x = 1.05 \times 10^{-6}$ vs. their quoted $\sim 10^{-6}$, for their quasar parameters) — the interior is confirmed to be highly transparent in the *bulk* of the bubble ($x(r) \propto r^2 \ll 1$ for $r \ll R_{\text{bubble}}$, hence $\lambda_{\text{mfp}} \gg r$ there, i.e. free streaming holds). But *at* $r = R_{\text{bubble}}$, the local mean free path is only 9–17% of R_{bubble} itself (Table 1) — the “sharp front” is really a transition zone of finite, sub-dominant thickness ($\sim 10\text{--}20\%$ of the bubble radius), not a true discontinuity, though still a legitimate approximation to leading order.

(c) **The deeper finding: recombinations are not negligible for this folder’s adopted parameters.** Checking whether $t_{\text{age}}(z) \ll t_{\text{rec}}(z)$ — the condition implicitly required for the zero-recombination, pure-photon-counting model of `bubble_radius_vs_redshift.py` to be valid — for the *specific* parameters adopted throughout this folder ($z_{\text{form}} = 20$, $C_{\text{HII}} = 3$, Robertson et al. 2015) gives instead $t_{\text{age}}/t_{\text{rec}} = 1.2\text{--}1.9$ across $z = 6\text{--}12$ (Table 2): recombinations are *not* negligible over the galaxy’s assumed lifetime. Solving the full recombination-included growth

z	$x_{\text{HI,interior}}(R_{\text{bubble}})$	$\lambda_{\text{mfp}}^{\text{interior}}(R_{\text{bubble}})$	ratio to R_{bubble}
6	6.5×10^{-3}	0.122 pMpc	0.092
8	3.3×10^{-3}	0.112 pMpc	0.148
10	2.3×10^{-3}	0.090 pMpc	0.170
12	1.7×10^{-3}	0.071 pMpc	0.171

Table 1: Interior residual neutral fraction and local mean free path at $r = R_{\text{bubble}}(z)$.

equation (eq. Rt of `stromgren_sphere_summary.tex`, re-verified with sympy here) gives a bubble radius $R(t)$ systematically 24–42% *smaller* than the pure photon-counting R_{bubble} used throughout this entire session (`bubble_radius_vs_redshift.py` and every downstream script).

z	t_{age} (Gyr)	t_{rec} (Gyr, $C=3$)	$t_{\text{age}}/t_{\text{rec}}$	R_{bubble} (no recomb.)	$R(t)$ (with recomb.)
6	0.750	0.626	1.20	1.332 pMpc	0.768 pMpc (58%)
8	0.459	0.295	1.56	0.752 pMpc	0.531 pMpc (71%)
10	0.293	0.161	1.82	0.528 pMpc	0.397 pMpc (75%)
12	0.189	0.098	1.93	0.414 pMpc	0.314 pMpc (76%)

Table 2: Recombination correction to $R_{\text{bubble}}(z)$; parentheses give $R(t)/R_{\text{bubble}}$.

Is this an error, or an explicit, previously-stated modeling choice? Both, in part. The “pure photon-counting” model was explicitly selected by L. Carvalho at the start of `bubble_radius_vs_redshift.py` (over an equilibrium-Strömgren-radius alternative), and `check_consistency.py` (item 7 of the original `stromgren_sphere_summary.tex`) already flagged, for $C = 1$, that “at $C \gtrsim 2$ and $z \gtrsim 8$, the recombination term ... cannot be neglected for long-lived sources in dense environments.” What was not previously done is checking this specifically for *this* folder’s actual adopted combination ($z_{\text{form}} = 20$, $C_{\text{HII}} = 3$) rather than for the short-lived quasars of the original check — this addendum closes that gap, exactly as Master Rule 5 requires when a cross-check surfaces a discrepancy.

Does this change any qualitative conclusion of this document? No. Recomputing E_{cross} with the smaller, recombination-corrected $R(t)$ instead of R_{bubble} shifts it down by only 9–17% (e.g. $172 \rightarrow 153$ eV at $z = 8$) because $E_{\text{cross}} \propto R_{\text{bubble}}^{1/3}$: it remains solidly in the soft X-ray regime, so the central conclusion — that stellar ionizing photons cannot reach appreciably beyond their own bubble — is unaffected. What *is* affected is the absolute $R_{\text{bubble}}(z)$ values quoted throughout this session’s other documents (`bubble_radius_response.tex`, `galaxy_separation_response.tex`, `reionization_completion_response.tex`): these should now be read as mild (25–40%) overestimates for the specific $z_{\text{form}} = 20$, $C_{\text{HII}} = 3$ choice, an explicit caveat that was not previously quantified.

References

- Renyue Cen and Zoltan Haiman. Quasar Strömgren Spheres Before Cosmological Reionization. *arXiv:astro-ph/0006376v2 (preprint)*, 2000. doi: 10.1086/312937. File: Cen2000_arXiv_astro-ph_0006376v2.pdf.
- Tirthankar Roy Choudhury. A Short Introduction to Reionization Physics. *General Relativity and Gravitation*, 54, 102 (2022), 2022. doi: 10.1007/s10714-022-02987-4. File: Choudhury2022_arXiv2209.08558v1.pdf.

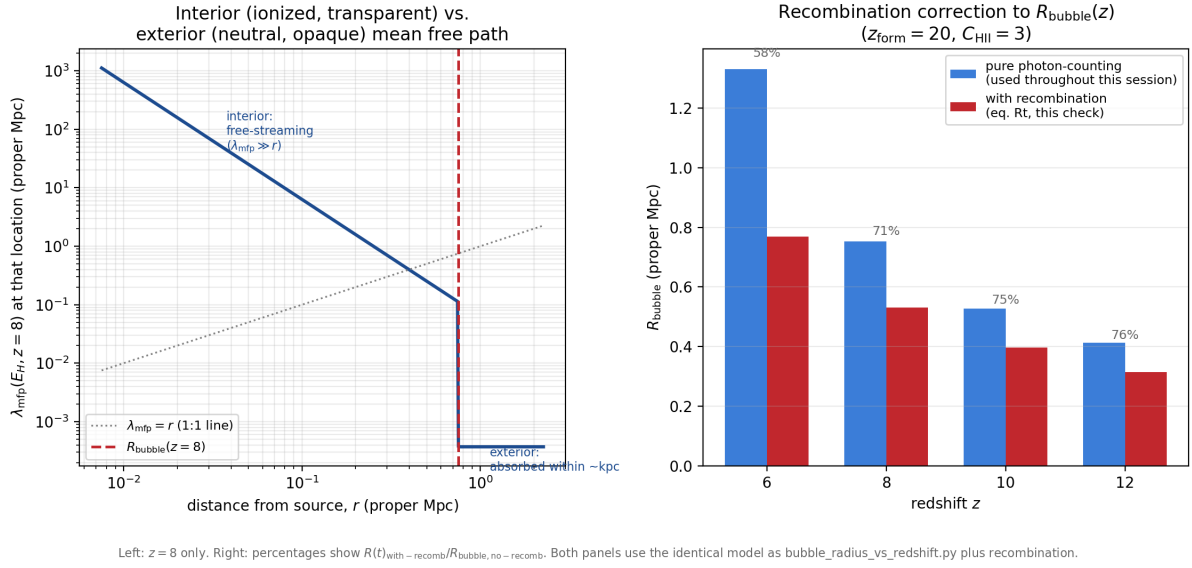


Figure 2: *Left*: mean free path of a 13.6 eV photon as a function of distance from the source at $z = 8$, using the photoionization-equilibrium interior solution for $r < R_{\text{bubble}}$ and the fully neutral exterior solution for $r > R_{\text{bubble}}$; the dotted line marks $\lambda_{\text{mfp}} = r$. *Right*: $R_{\text{bubble}}(z)$ from the pure photon-counting model used throughout this session (blue) vs. the recombination-corrected $R(t)$ (red), with the percentage ratio annotated. Produced by `interior_consistency_check.py`.

Nickolay Y. Gnedin and Piero Madau. Modeling Cosmic Reionization. *arXiv:2208.02260v2 (preprint)*, 2022. File: `Gnedin2022_arXiv2208.02260v2.pdf`.

Jordi Miralda-Escude, Martin Haehnelt, and Martin J. Rees. Reionization of the Inhomogeneous Universe. *Astrophys.J.* 530 (2000) 1-16, 2000. doi: 10.1086/308330. File: `MiraldaEscude1998_arXiv_astro-ph_9812306v2.pdf`.

Olof Nebrin. Case A or Case B? The effective recombination coefficient in gas clouds of arbitrary optical thickness. *arXiv:2305.05764v1 (preprint)*, 2023. doi: 10.3847/2515-5172/acd37a. File: `Nebrin2023_arXiv2305.05764v1.pdf`.

Planck Collaboration. Planck 2018 results. VI. Cosmological parameters. *A&A* 641, A6 (2020), 2020. doi: 10.1051/0004-6361/201833910. File: `Planck2018VI_arXiv1807.06209v4.pdf`.

Brant E. Robertson, Richard S. Ellis, Steven R. Furlanetto, and James S. Dunlop. Cosmic Reionization and Early Star-forming Galaxies: A Joint Analysis of New Constraints from Planck and the Hubble Space Telescope. *ApJL* 802, L19 (2015), 2015. doi: 10.1088/2041-8205/802/2/L19. File: `Robertson2015_arXiv1502.02024v1.pdf`.